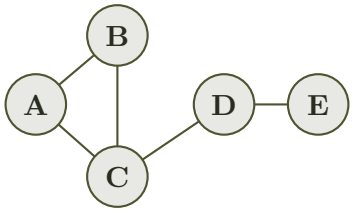
Module 8: Graph Theory and Networks — Prequiz

Open-note practice. Show your counts and routes. Brief explanations are enough.

Name: _____ Date: _____



Use the graph above for all five questions. Edges: AB, AC, BC, CD, DE.

1. What are C's direct neighbors? Find the degree of C. Check the handshake rule by adding all five degrees.

2. Find one shortest path from A to E. Write its nodes in order and give its length in edges. Explain why A and E are connected even though no direct edge joins them.

3. Suppose C–D fails. List the connected components. Is E reachable from A? Explain briefly.

4. The town can add just one connection: A–E or B–D. Both choices preserve a connection between A and E after C–D fails. Trace a surviving path for each choice. Which one gives the shorter A-to-E path?

5. Begin with only A informed in round 0. Every informed node alerts its direct neighbors in the next round. Newly informed nodes relay one round later. List the **newly informed nodes** in rounds 1, 2, and 3 and the **total number informed** after round 2.